



AWS CLOUD COMPUTING

CAPSTONE PROJECT 09.12.2025

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AGENDA AND GOAL OF THE PROJECT

- (1) Understanding the Project Context
- (2) Method – Terraform Deployment Workflow
- (3) Requirements and Architectural Design of the Advanced Project
- (4) Key Challenges
- (5) Final Results and Conclusions



Deployment of a scalable Wordpress Website with a MySQL database on AWS (using Terraform)



Creating a travel blog about a Cycladic island Amorgos to showcase local highlights, culture, and travel tips

THE JOURNEY TO THE CLOUD: MOTIVATION & AWS BENEFITS



- Hands-on experience with modern cloud and Infrastructure-as-Code practices.
- Realistic use case with practical value
- Combines technical implementation with creative content



Scalability

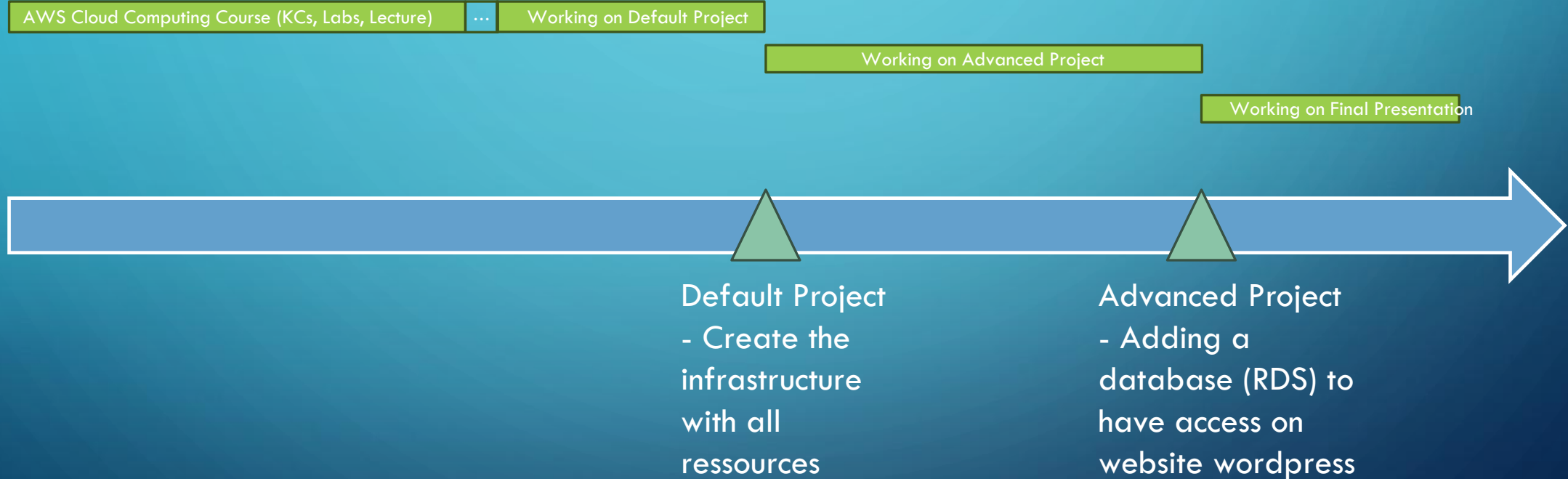
Reliability & Availability

**Automation with
Terraform**

Security

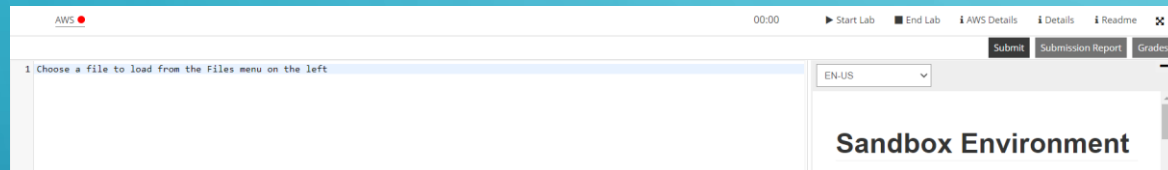
Cost Efficiency

WHAT COULD I ACHIEVED DURING THE PROJECT PHASE?

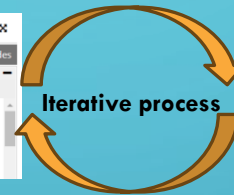


METHOD AND WORKFLOW

First Steps in AWS Management Console (Using Sandbox Environment)



Instructions for Project Steps



Start with IaS
aws configure

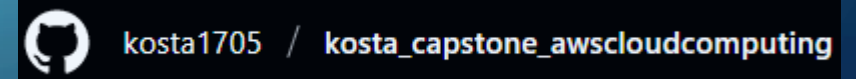
Local via CLI:

- terraform init
- terraform plan
- terraform apply

Creation of .tf files



Push to my GitHub repository

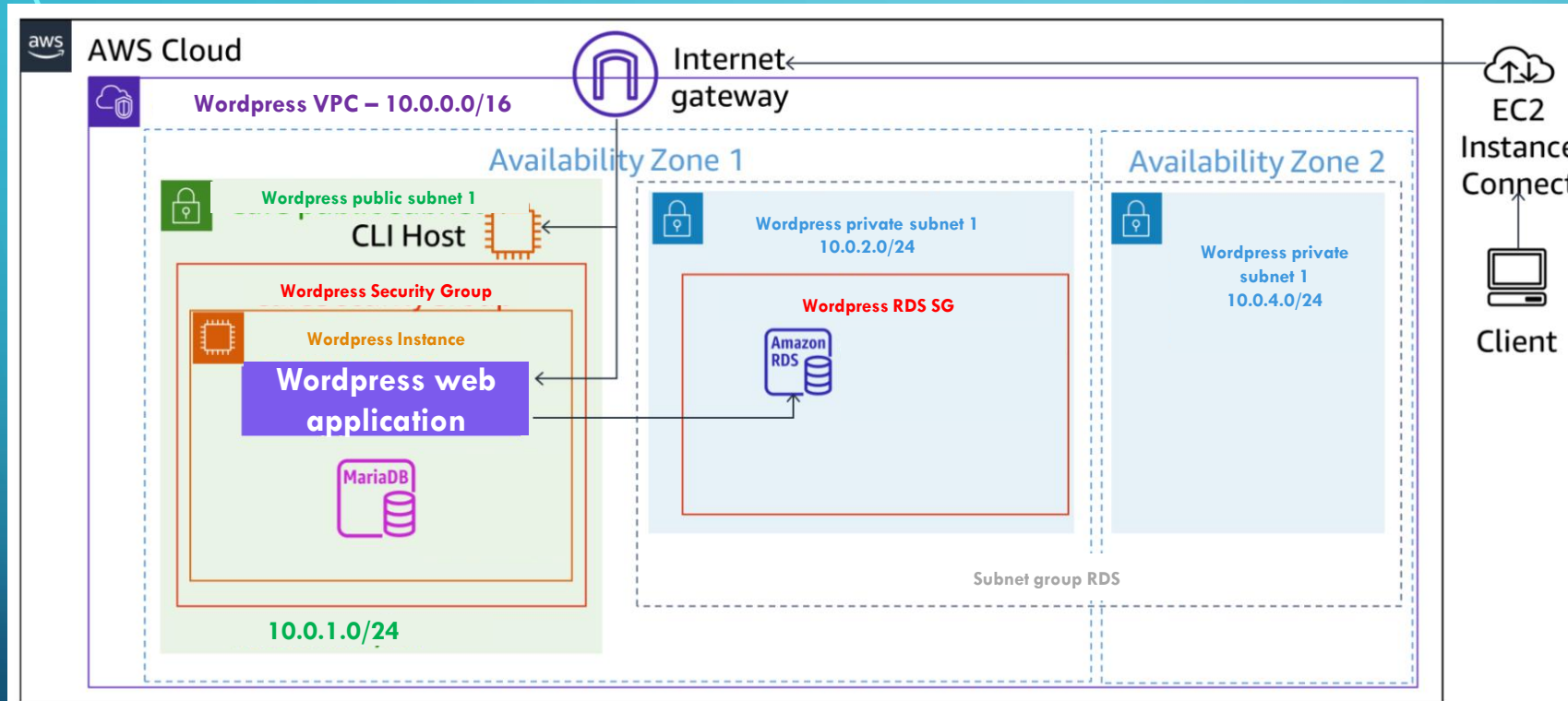


https://github.com/kosta1705/kosta_capstone_awscloudcomputing/tree/main

WHICH TECHNICAL REQUIREMENTS NEED TO BE FULFILLED BY THE ARCHITECTURE?

Default Project (Level 1)	Advanced Project (Level 2)
R1: IaS → using VS Code and Terraform	R1: Adding RDS (access with private subnet?) – database will be on RDS
R2: Resources • 2 AZ (Public and Private Subnet) • IGW • EC2 Instance (Amazon Linux 2) • Security Groups (SSH and HTTP) • Local database to EC2	R2: Adaption of Security Groups (due to RDS with Port 3306 / Protocol TCP)
R3: User Data Script (Wordpress instruction)	R3: Adjustment of User Data Script in Terraform

ARCHITECTURE WITH ADVANCED PROJECT STATUS



Components as .tf-files:

- EC2
- VPC
- Provider (region)
- Internet Gateway
- Public and private subnets
- Security group
- Variables
- Route Table and Association
- RDS (+Subnet Group, +Security Group)
- Outputs

+

userdata.sh

+

terraform.tfvars

WHICH CHALLENGES DID I HAVE TO SOLVE ALONG THE WAY?

Handling security groups
and database access



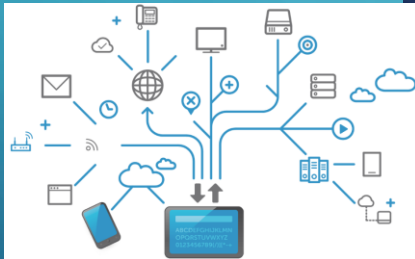
<https://www.sysdig.com/blog/aws-security-groups-guide/>

Modularization



<https://wuepping.com/seminar-modularisierung/>

Proper Networking



<https://assignmentstudio.net/what-are-networks/>

Debugging user data scripts



<https://www.wilde-it.com/debugging/>



LESSONS LEARNED AT A GLANCE

- ✓ Successful implementation of a scalable Wordpress environment with AWS resources using Terraform
- ✓ Gained understanding and hands-on experience with key cloud components such as VPC, subnets, EC2, security groups, and RDS
- ✓ Improved/Upgraded architecture by separating webserver and database and using Multi-AZ for increased availability
- ✓ Created modular, reproducible, and version-controlled infrastructure code using Terraform
- ✓ Improvements: WordPress Servers to private subnet, Autoscaling, Application Load Balancer, CloudWatch monitoring and alert, ...

Instances (1) [Info](#)

Connect Instance state ▾ Actions ▾ Launch instances

All states ▾ < 1 > ⚙️

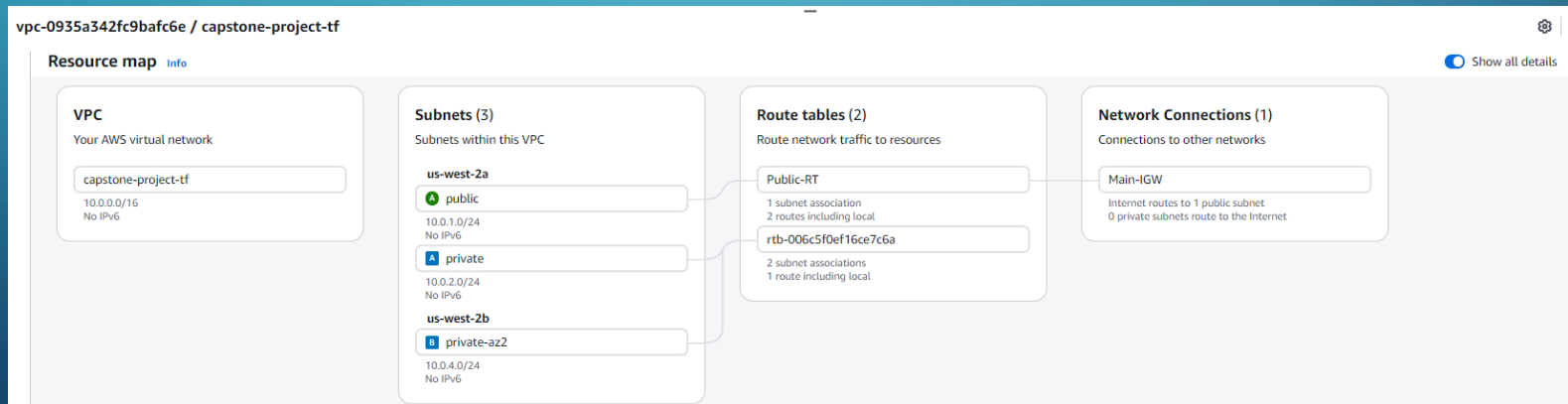
☐	Name ↗	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status	Availability
☐	Capstone-tf-Instance	i-0424d8330fb9d2a8b	Running 🔍	t3.small	3/3 checks passed 🔍	View alarms +	us-west-2a

Databases (1)

Group resources Modify Actions ▾ Create database

< 1 > ⚙️

☐	DB identifier ↗	Status ▾	Role ▾	Engine ▾	Upgrade rollout order ▾	Region ... ▾
☐	wordpress-db	Available 🔍	Instance	MariaDB	SECOND	us-west-2b



AMORGOS TRAVEL DIARY: HIDDEN PATHS, DEEP BLUE, TRUE GREECE

← → ↻ ⚠ Nicht sicher | http://35.91.18.139/index.php/2025/12/05/hello-world/

WordPress Capstone Project Edit Site 0 + New Edit Post

amorgos.



Written by [Kopapa](#) in [Uncategorized](#)

The Monastery of Panagia Hozoviotissa

Built into a towering cliff 300 meters above the sea, this iconic 11th-century monastery is one of the most breathtaking sights in all of the Cyclades. Climbing the steep white steps is challenging, but the reward—a stunning view over the endless blue of the Aegean—is unforgettable.